

Milestone-4 (Incident Type Analysis of US Disaster Declarations)

1. Introduction

Natural disasters occur in different forms such as floods, storms, and hurricanes. Understanding the distribution of disaster types is essential to analyze their frequency, impact, and regional variation.

In this milestone, an in-depth analysis of **incident types** is performed. The analysis also integrates insights from previous milestones (temporal and geographical analysis) to provide a complete understanding of disaster patterns.

The focus of this milestone is to identify:

- Most frequent disaster types
- Distribution of disaster types across states
- Relationship between disaster types and assistance programs
- Overall insights combining time, location, and disaster categories

2. Dataset Description

The dataset used in this milestone is the cleaned dataset from previous milestones along with additional data from the original dataset.

Dataset Used

- usnd_cleaned.csv (processed dataset)
- us_disaster_declarations.csv (original dataset for assistance columns)

Columns Used

Column	Description
state	State where disaster occurred
incidentType	Type of disaster
year	Year of occurrence
month	Month of occurrence
ih_program_declared	Individual housing assistance
pa_program_declared	Public assistance

3. Data Preparation

To perform assistance analysis, the cleaned dataset was merged with the original dataset.

Method Used: Merge Operation

The **merge()** function in Pandas is used to combine two datasets based on common columns.

```
pd.merge(clean_df, raw_df, on=['state','incidentType'])
```

Theory

Merge operation joins datasets using common keys.

This allows combining cleaned features with additional attributes such as assistance programs.

4. Incident Type Distribution Analysis

Method Used: value_counts()

The frequency of each disaster type was calculated using:

```
df['incidentType'].value_counts()
```

Theory

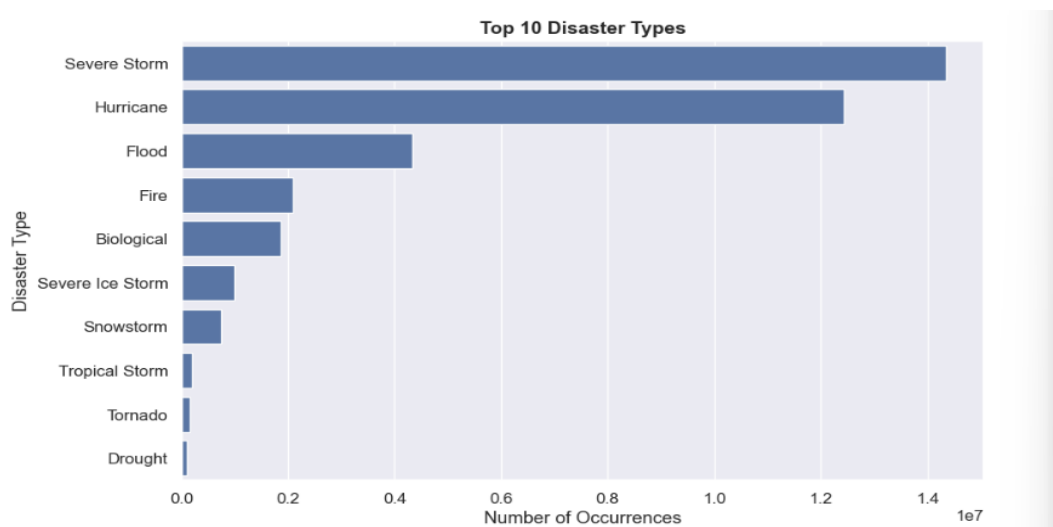
value_counts() counts occurrences of each category in a column.

Visualization: Bar Chart

A bar chart was used to display the **top 10 disaster types**.

Theory of Bar Chart

A bar chart represents categorical data using rectangular bars, where the height or length corresponds to the value.



Interpretation

The visualization shows that **Severe Storms, Hurricanes, and Floods** are the most frequent disaster types.

This indicates that **weather-related disasters dominate across the United States**.

5. State vs Incident Type Analysis

Method Used: groupby()

```
df.groupby(['state','incidentType']).size()
```

Theory

groupby() groups data based on multiple columns and applies aggregation functions.

Pivot Table

```
pivot_data = incident_state.pivot(index='state', columns='incidentType', values='count')
```

Theory of Pivot Table

A pivot table restructures data:

- Rows → states
 - Columns → disaster types
 - Values → counts
-

6. Heatmap Visualization

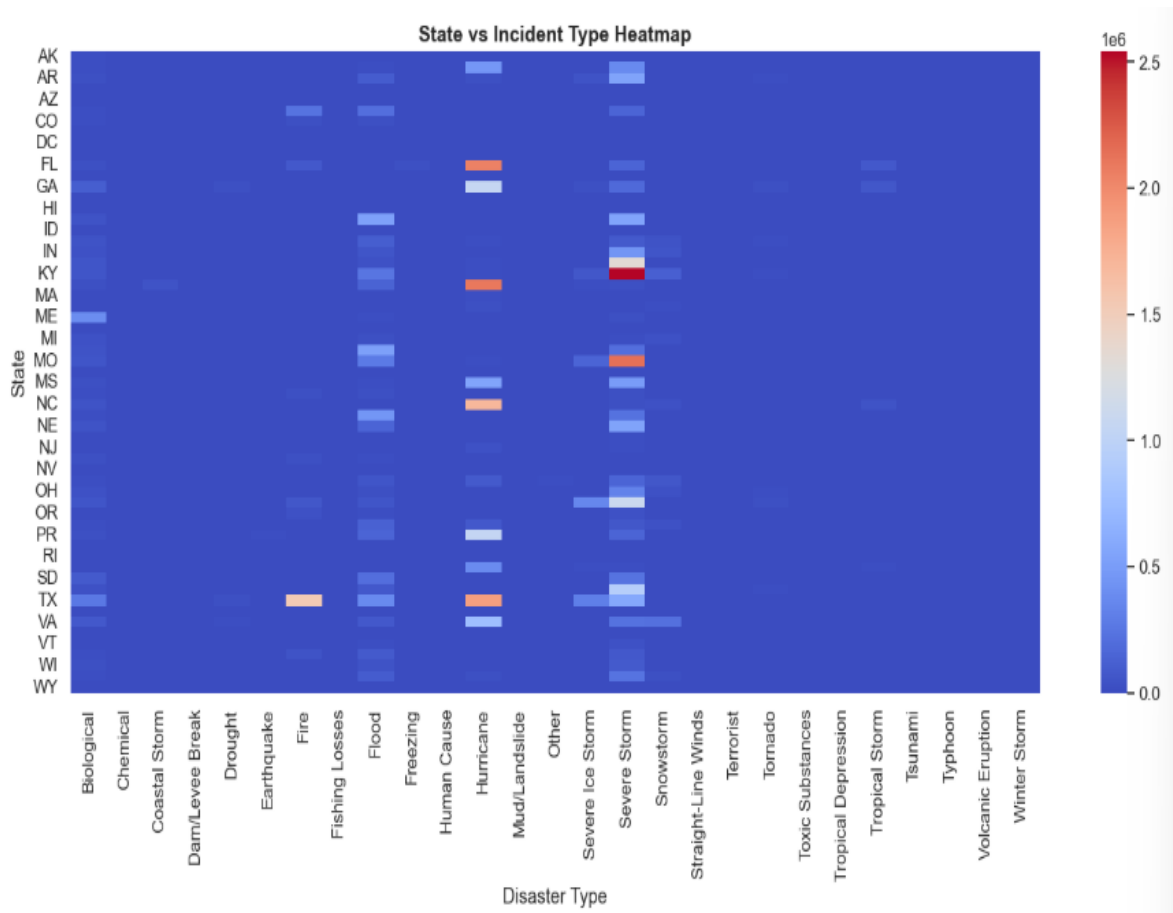
Visualization: Heatmap

A heatmap was used to visualize disaster distribution across states.

Theory of Heatmap

A heatmap uses color intensity to represent values in a matrix format.

- Darker colors → higher values
- Lighter colors → lower values



Interpretation

The heatmap shows that different states experience different disaster types.

- Coastal states have more hurricanes
- Other regions show higher flood or storm activity

This indicates strong **geographical influence on disaster types**.

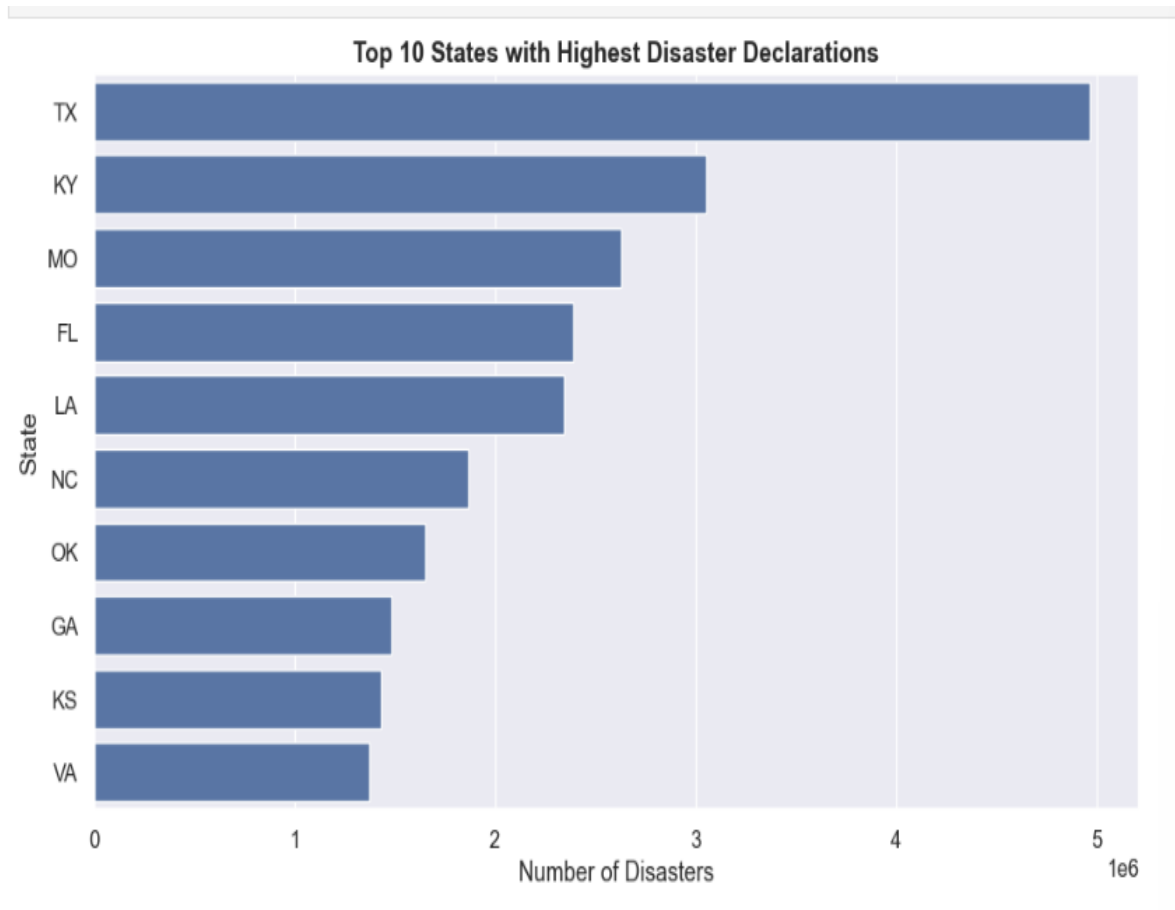
8. State-Level Disaster Comparison

Method Used: Aggregation

```
df.groupby('state').size()
```

Visualization: Bar Chart

Top 10 states with highest disaster declarations were visualized.



Interpretation

States such as **Texas, California, and Florida** have the highest disaster occurrences.

This is due to their geographic size and exposure to various disaster types.

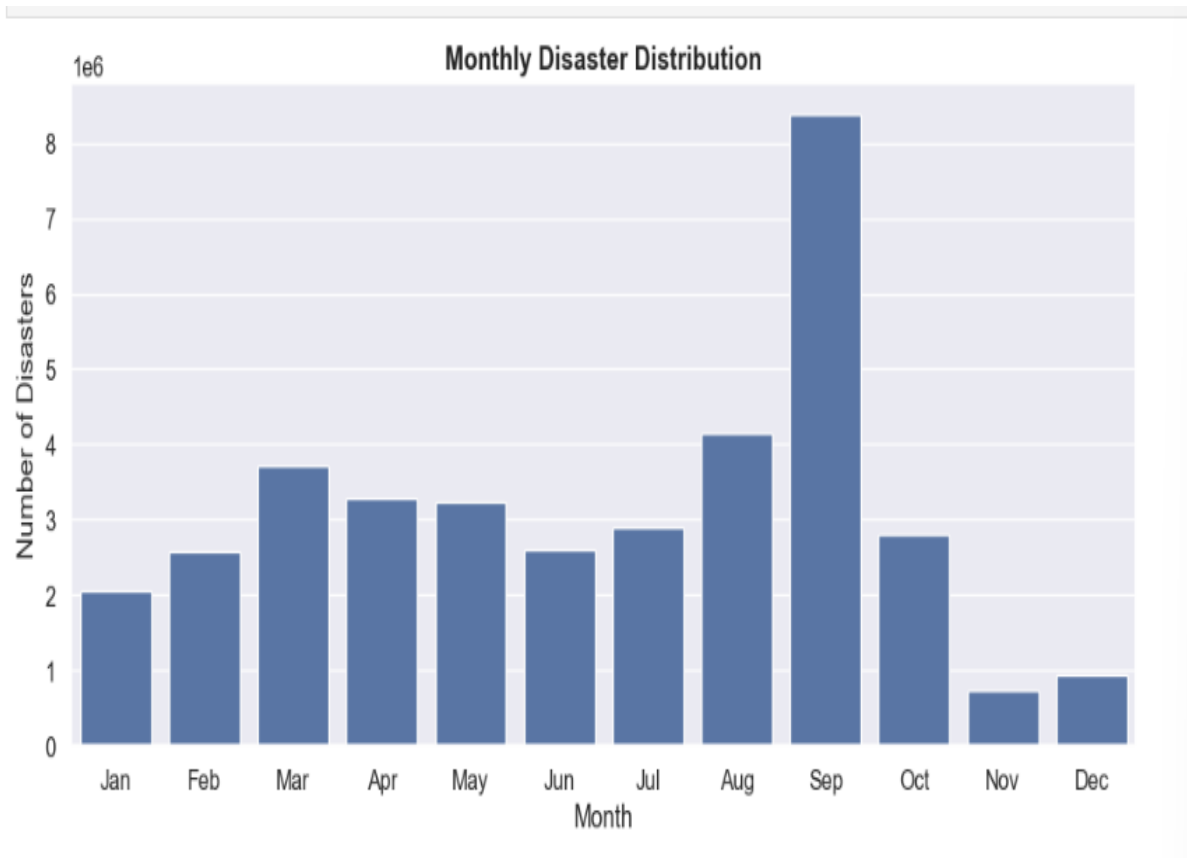
9. Monthly Distribution Analysis

Method Used

```
df.groupby('month').size()
```

Visualization: Bar Chart

Monthly disaster frequency was visualized.



Interpretation

Disasters show seasonal patterns, with certain months having higher frequency.

This reflects **climate-related seasonal effects such as hurricanes and rainfall cycles.**

10. Assistance Program Analysis

Method Used

```
df.groupby('incidentType')[['ih_program_declared','pa_program_declared']].sum()
```

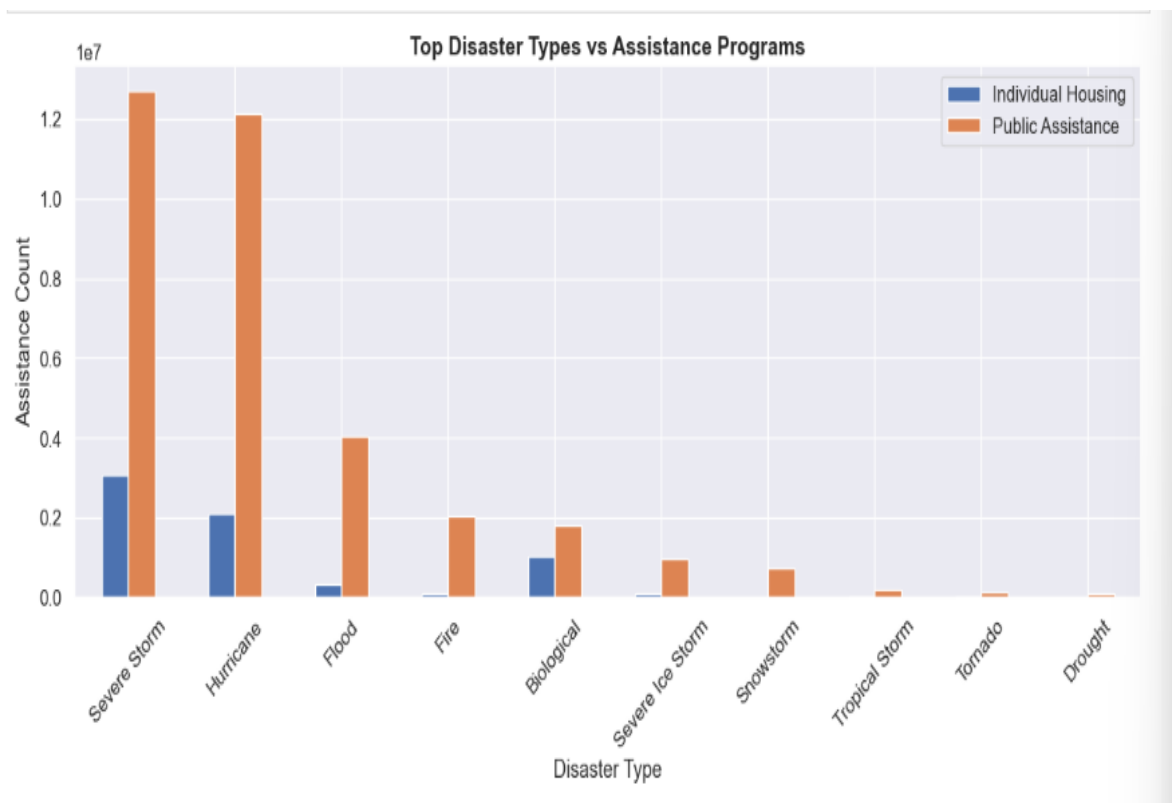
Theory

This aggregates the total number of times assistance programs were activated for each disaster type.

Visualization: Grouped Bar Chart

A grouped bar chart compares:

- Individual housing assistance
- Public assistance



Interpretation

Disasters such as hurricanes and floods trigger higher assistance programs.

Public assistance is more frequently used, indicating large-scale disaster impact.

11. Final Integrated Insights

This milestone combines findings from all previous analyses.

- Disaster frequency is increasing over time
- Certain states are more disaster-prone
- Weather-related disasters dominate
- Disaster type varies by region
- Assistance programs are linked to high-impact disasters

12. Conclusion

In this milestone, a detailed analysis of disaster incident types was performed.

The study revealed that disaster patterns depend on multiple factors including time, location, and type of disaster.

The integration of all milestones provides a complete understanding of disaster trends and helps in identifying risk-prone regions and disaster types.

13. Deployed Dashboard Link

The interactive dashboard for this project has been successfully deployed and can be accessed using the link below:

Live Dashboard: <https://usnaturaldisaster.vercel.app/>